

System Architecture Design

Project Title: **DeFi Staking Platform**

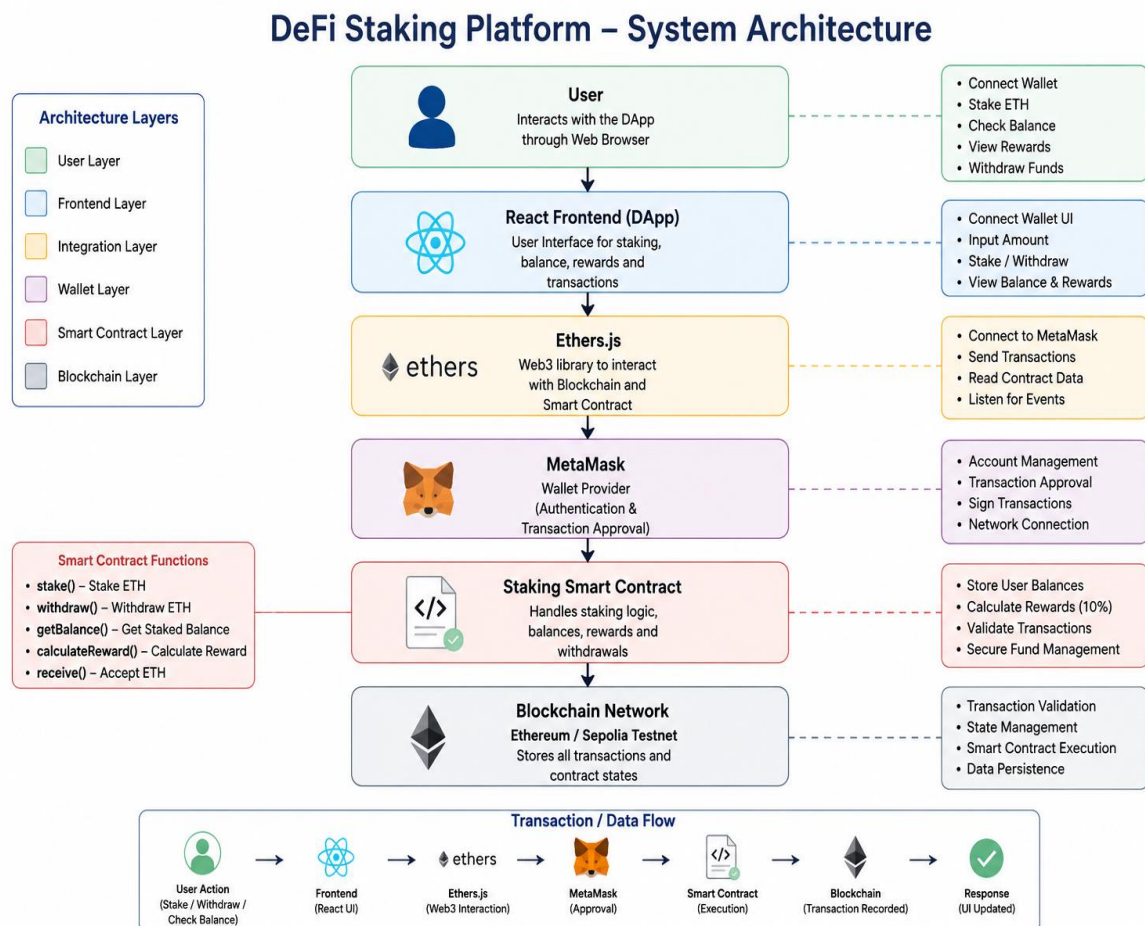
System Overview

The DeFi Staking Platform is a decentralized application (DApp) that allows users to stake ETH, earn rewards, view balances, and withdraw funds securely using blockchain technology.

The platform consists of the following layers:

1. User Layer
2. Frontend Layer (React.js)
3. Integration Layer (Ethers.js)
4. Wallet Layer (MetaMask)
5. Smart Contract Layer (Solidity)
6. Blockchain Layer (SCAI Mainnet)

System Architecture



Component Description

User Layer

The user interacts with the application through a web browser.

Responsibilities

- Connect wallet
- Stake ETH
- Check balance
- View rewards
- Withdraw funds

Frontend Layer

The frontend is built using React.js and Vite.

Responsibilities

- Display staking dashboard
- Accept user input
- Show balances and rewards
- Communicate with smart contract

Technologies

- React.js
- Vite
- JavaScript

Integration Layer

Ethers.js connects the frontend to the blockchain.

Responsibilities

- Read blockchain data
- Send transactions
- Call smart contract functions

Technology

- Ethers.js v6

Wallet Layer

MetaMask is used for authentication and transaction approval.

Responsibilities

- Wallet connection
- Transaction signing
- Account management

Smart Contract Layer

The staking contract is written in Solidity.

Functions

stake()

Allows users to stake ETH.

withdraw()

Allows users to withdraw staked ETH.

getBalance()

Returns user staking balance.

calculateReward()

Calculates staking reward as 10% of the staked amount.

claimReward()

Returns the estimated reward amount.

Blockchain Layer

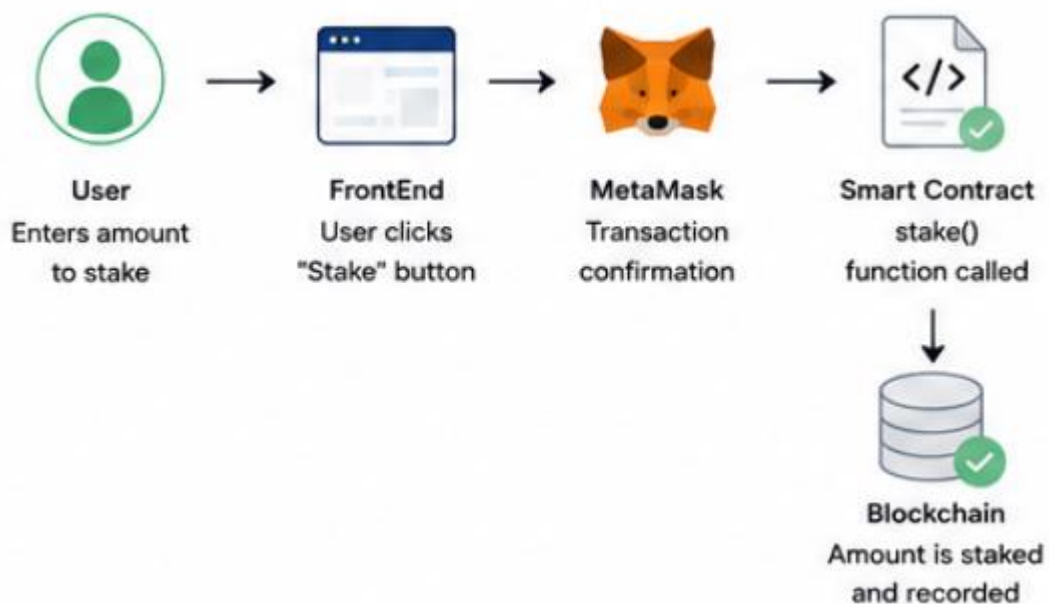
The deployed smart contract runs on the SCAI Mainnet.

Responsibilities

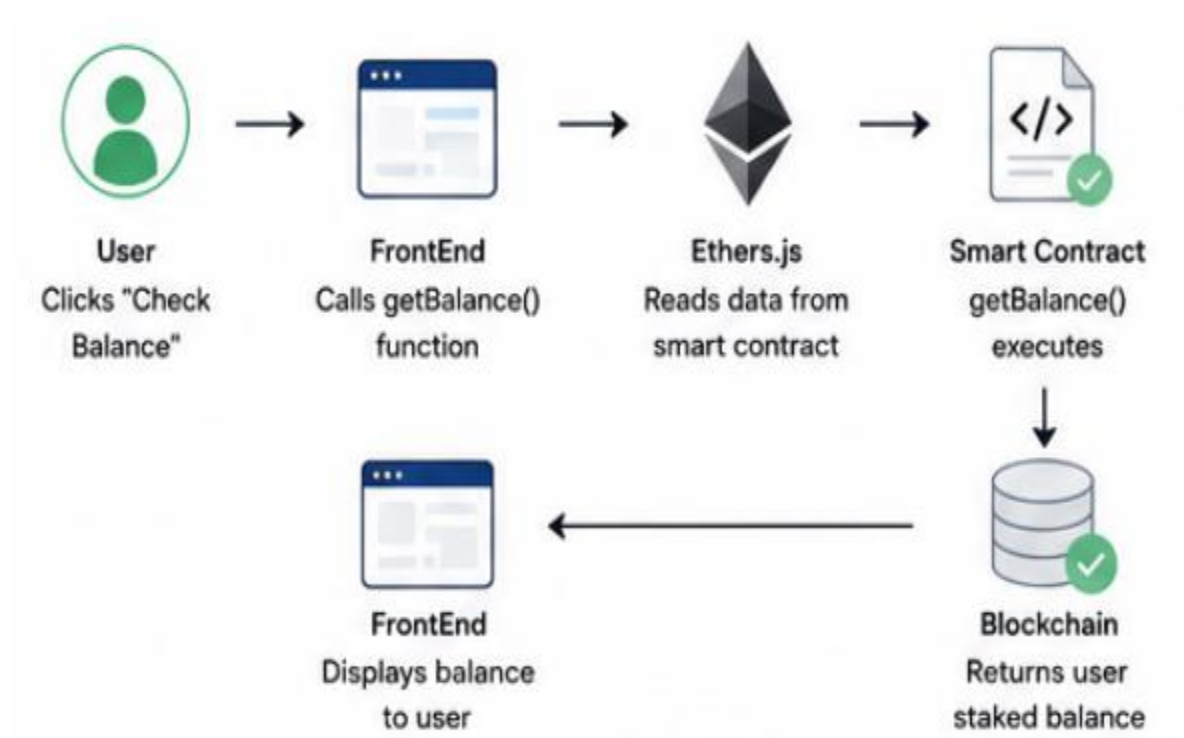
- Execute smart contracts
- Validate transactions
- Store blockchain state

Smart Contract Logic

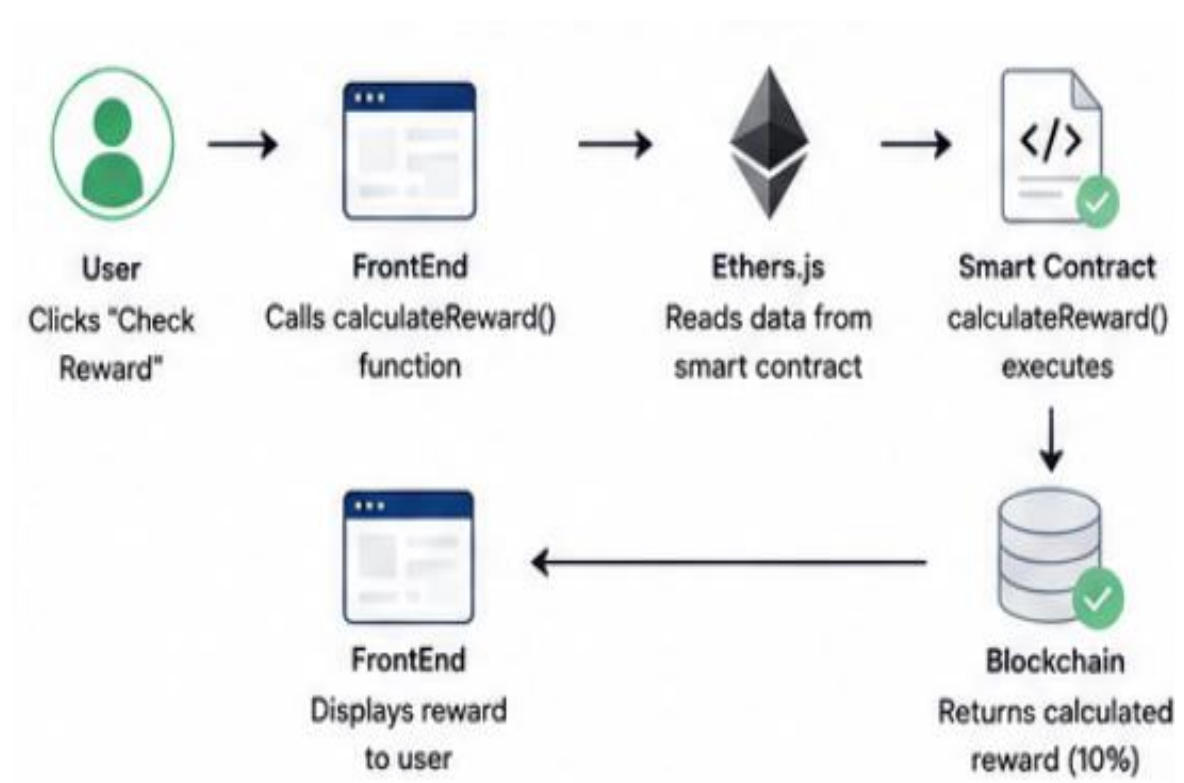
Stake Workflow



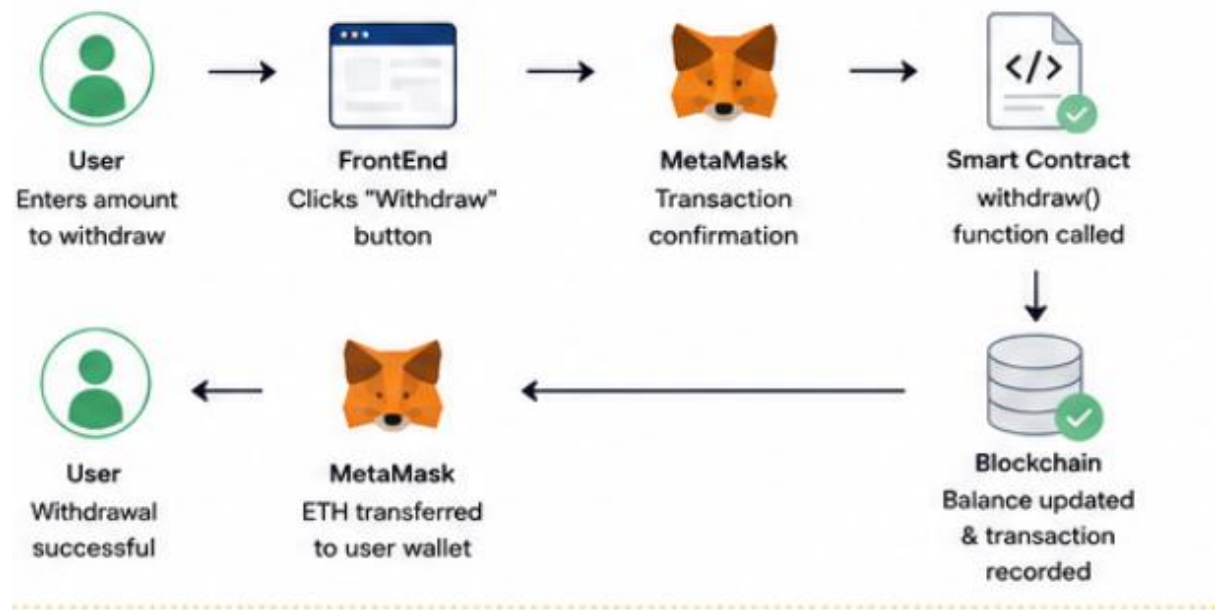
Balance Workflow



Reward Workflow



Withdrawal Workflow



Data Flow



Security Considerations

- MetaMask wallet authentication
- Input validation using `require()`
- Balance verification before withdrawal
- Solidity 0.8.x overflow protection
- Transparent blockchain transactions

Conclusion

The DeFi Staking Platform follows a layered architecture using React.js, Ethers.js, MetaMask, Solidity, and blockchain technology. The architecture ensures secure staking, reward calculation, balance tracking, and withdrawal operations through decentralized smart contracts.